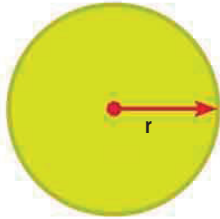


## Area

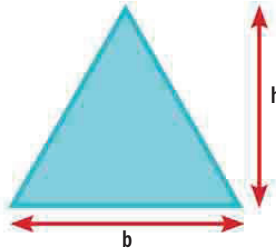
The area of a shape is the amount of space inside it. Formulas for working out the areas of common shapes are given below.



$$\text{area} = \pi r^2$$

### △ Circle

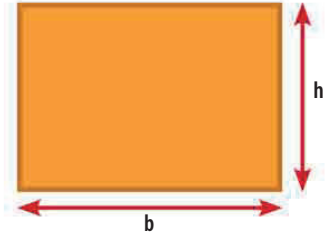
The area of a circle equals pi ( $\pi = 3.14$ ) multiplied by the square of its radius.



$$\text{area} = \frac{1}{2}bh$$

### △ Triangle

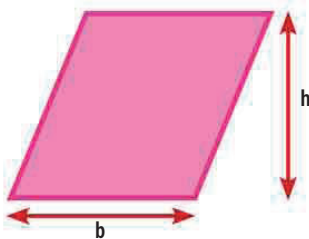
The area of a triangle equals half multiplied by its base multiplied by its vertical height.



$$\text{area} = bh$$

### △ Rectangle

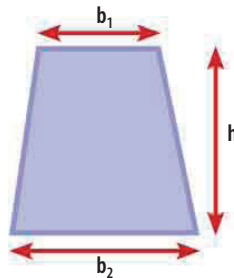
The area of a rectangle equals its base multiplied by its height.



$$\text{area} = bh$$

### △ Parallelogram

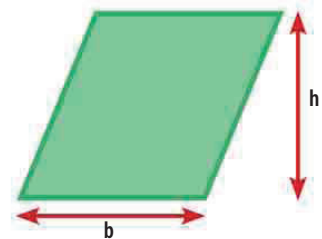
The area of a parallelogram equals its base multiplied by its vertical height.



$$\text{area} = \frac{1}{2}h(b_1 + b_2)$$

### △ Trapezoid

The area of a trapezoid equals the sum of the two parallel sides, multiplied by the vertical height, then multiplied by  $\frac{1}{2}$ .



$$\text{area} = bh$$

### △ Rhombus

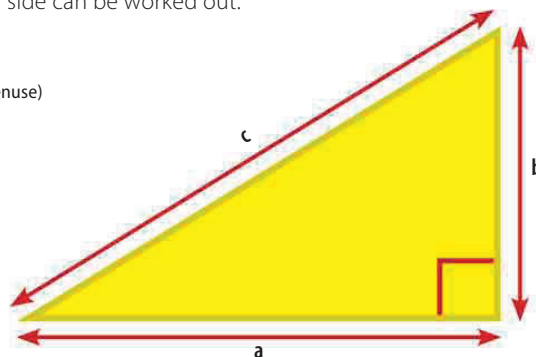
The area of a rhombus equals its base multiplied by its vertical height.

## Pythagorean Theorem

This theorem relates the lengths of all the sides of a right triangle, so that if any two sides are known, the length of the third side can be worked out.

$$a^2 + b^2 = c^2$$

side a  
side b  
side c (hypotenuse)



### ◁ The theorem

In a right triangle the square of the hypotenuse (the largest side,  $c$ ) is the sum of the squares of the other two sides ( $a$  and  $b$ ).